ANAIS 2026 Application Responses

Applicant: Saransha Basu

1. Academic and Professional Background

I am pursuing a Bachelor in Computer Engineering at Khwopa Engineering College, Bhaktapur, Nepal, and expect to graduate in 2027. After completing four semesters, my semester GPAs are 3.56, 3.58, 3.64, and 3.71. Because the semesters had equal credits, my cumulative GPA is approximately 3.62 out of 4.00. I have not yet held a formal internship or full-time professional position. My current focus is on building practical skills in machine learning, deep learning, natural language processing, computer vision, and backend development. Alongside my coursework, I regularly implement algorithms and neural-network concepts from first principles and document my learning through projects and technical notes. My public project portfolio includes work in Python, NumPy, PyTorch, FastAPI, NLP, semantic search, and audio classification.

2. Motivation and Expected Benefits

I want to apply to ANAIS because I have developed a strong theoretical interest in artificial intelligence but need more experience applying that knowledge to challenging, real-world problems. I have studied several books and created personal notes on machine learning and deep learning, but an intensive school would help me connect those ideas with better experimentation, implementation, and research practice. I hope to learn from instructors and participants working on serious AI problems, receive feedback on my approach, and understand how to design more reliable and meaningful projects. ANAIS would support my academic development, help me identify research directions, and strengthen my preparation for future work in machine learning and deep learning. I also hope to contribute by sharing my first-principles understanding, practical learning notes, and experience mentoring classmates.

3. Mathematical Background and Programming Proficiency

My engineering mathematics courses covered calculus, probability, linear algebra, discrete mathematics, statistics, and numerical methods. I have used calculus, probability, and linear algebra while implementing machine-learning algorithms from scratch and while studying optimization, gradients, model training, and data representations. Python is my primary programming language for machine learning and data work, and I use NumPy, Pandas, Jupyter, PyTorch, TensorFlow, Keras, and scikit-learn. I also have practical experience with C and C++ through data structures, algorithms, and computer-graphics coursework, as well as JavaScript and TypeScript for web applications. I have implemented linear regression, logistic regression, decision trees, K-means clustering, PCA, gradient descent, regularization techniques, and an automatic-differentiation engine. These projects have helped me understand not only how to use libraries, but also how the underlying mathematical and computational mechanisms work.

4. Relevant Technical or Research Projects

The project I am most proud of is an automatic-differentiation engine implemented in pure NumPy. I built dynamic computation graphs, tracked parent-child relationships, accumulated gradients, and implemented reverse-mode backpropagation for operations such as addition, multiplication, matrix multiplication, reductions, broadcasting, and transposition. I then used the engine to train a two-layer neural network on XOR, which helped me understand mechanisms normally hidden inside deep-learning frameworks. I also worked on BIRDCLEF 2025, where I compared a conventional CNN, a hybrid CNN with attention, and an Audio Spectrogram Transformer for bird-audio classification. In

addition, I completed Semantic-Note, a FastAPI application using sentence embeddings for semantic note search, and maintain a portfolio of 50 focused projects for understanding LLM concepts. Other work includes an RNN question-answering system, a next-word predictor, and classical ML algorithms implemented from scratch.

5. Extracurricular and Community Activities

I have participated in three hackathons: the Himalaya College Hackathon, DAV Hackathon, and ACME Hackathon. I have also attended GDG developer events to learn from the technology community and interact with other developers. Outside formal coursework, I maintain a consistent self-directed learning routine through technical books and implementation projects. Books that have influenced my learning include Hands-On Machine Learning, 50 ML Projects to Understand LLMs, Grokking Deep Learning, Understanding Deep Learning, Clean Code, and Clean Architecture. I also mentor classmates in machine-learning fundamentals, deep-learning fundamentals, Python, NumPy, Pandas, and NLP, and I prepare structured notes to explain concepts from first principles.

6. Scholarship Rationale and Personal Circumstances

I am applying for financial support because I come from a middle-class family and do not have sufficient personal financial resources to comfortably cover the costs of participating in an international AI school. Despite this limitation, I have maintained strong academic performance, with a current cumulative GPA of approximately 3.62 out of 4.00, and I have consistently invested my own time in building practical machine-learning and deep-learning skills. I believe I would use the opportunity seriously and contribute actively to other participants. A scholarship would make it possible for me to access intensive training, expert guidance, and a diverse learning environment that would otherwise be difficult for me to afford. My academic record, sustained self-directed study, technical portfolio, and commitment to sharing knowledge demonstrate my merit and potential benefit from this support.

7. Additional Supporting Information or TA Application

I would like to be considered for a Teaching Assistant role, subject to the programme's eligibility requirements. I can confidently support participants with Python, NumPy, Pandas, machine-learning fundamentals, deep-learning fundamentals, NLP, and the first-principles implementation of algorithms and neural networks. I have mentored classmates and prepared more than 15 sets of personal notes on ML and DL theory. I could help participants understand mathematical intuition, translate theory into code, debug notebooks, and reason about model choices. My strongest supporting projects are the NumPy autograd engine, the BIRDCLEF model comparison, the Semantic-Note semantic-search system, and my LLM learning portfolio. I understand that the TA role may require publications or extensive industry experience; I do not claim those credentials. I am presenting my mentoring experience, technical projects, and ability to explain fundamentals honestly for consideration.

8. Generative AI Use Declaration

Yes. I used ChatGPT to help organize my ideas, improve grammar and clarity, calculate and present my cumulative GPA, and format my responses. The experiences, academic information, projects, motivations, and personal circumstances described in these answers are my own. I reviewed the wording and take responsibility for ensuring that the submitted information is accurate and authentic.